

# Natural History





## About the Course

Learners encounter botanical families, endangered species, invasive species, and secondary succession as they consider how man interacts with and thinks about the natural world. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 5

### About Science: Grade 5

In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

### Science: Grade 5

#### The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more "friends" in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge "by the way" and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



## Placement & Combining Tips

### About Science: Grade 5

In level 5, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from botany, astronomy, and more.

### Natural History: Grade 5

For approximately fifth-grade students based on a balance of complexity and reading level. However, learners may be freely combined, or sequence reordered based on needs and preferences.



## Scheduling

| GRADE | SCHEDULE INFO.                                 | BOOKS                                                                    |
|-------|------------------------------------------------|--------------------------------------------------------------------------|
| 5     | Natural History: Grade 5<br>1 time/week 15 min | Beetle Busters<br>Nature All Around: Plants<br>Shanley's Quest Card Game |

| GRADE | SCHEDULE INFO. | BOOKS                                        |
|-------|----------------|----------------------------------------------|
|       |                | The Bat Scientists<br>The Forest in the Tree |

#### Sample Weekly View

| Day 1                       | Day 2 | Day 3                                                                 | Day 4 | Day 5                                        |
|-----------------------------|-------|-----------------------------------------------------------------------|-------|----------------------------------------------|
| <b>Science: Grade 5</b>     |       |                                                                       |       |                                              |
| Natural History:<br>Grade 5 |       | General Science:<br>Grade 5<br>Nature Walks &<br>Scouting: Grades 1-8 |       | Labs: Grade 5<br>Nature Notebook:<br>Grade 5 |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 5

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

## Special Topics & Field Trips

Science: Grade 5

Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

Natural History: Grade 5

- ☐ Term 1: a botanical garden or plant conservatory
- ☐ Term 2: evening walks and/or a cave, a wildlife rehabilitator, or bat conservationist
- ☐ Term 3: a forestry museum or tour

## Term Prep & Teacher Tips

Science: Grade 5

Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

Natural History: Grade 5

- ☐ Term 1: one or more wildflowers from p.461-508
- ☐ Term 2: bats p.241-244
- ☐ Term 3: one or more trees from p.628-692, one or more insects from p.301-395

- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
  - flower pot with soil, or place in the ground
  - trowel
  - globe/wall map

## Reminders

Natural History: Grade 5

□ Note that term 1 of Natural History begins with planting a small garden, so be sure to have those supplies and any others indicated on the supply list ready. For indoor situations, the choices are many, provided there is a sunny window. For outdoors in a cooler season, some good options are parsley, carrots, or peas (may need a trellis). Basil, thyme, or beans (may need a trellis) are good options for outdoor planting in a warmer season. Note that some plants will not reach the flowering stage for a while.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Natural History: Grade 5



Beetle Busters



Nature All Around: Plants



Shanleya's Quest Card Game



The Bat Scientists



The Forest in the Tree



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Natural History: Grade 5



Household Items - Natural History: Grade 5



## Quick Links

### Related Course Quick Links

- ∞ [Extra Helpings](#)
- ∞ [Outdoor Work](#)
- ∞ [Seek app from iNaturalist](#)
- ∞ [SkyView Lite for iOS](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [Foundations \(See Section 13: Science\)](#)

Click THIS text  
or scan the QR  
code for links.



# Natural History: Grade 5

## How To Teach



### Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



### Read

- Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.



### Narrate

- Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.
- Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.
- Teachers may take turns to model.



### Discuss

- Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.
- Notice if there were any dates that they want to keep for their Book of Centuries.



## Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.
  - These can be viewed as alternative ways to engage.
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SAMPLE

# Natural History: Grade 5

[Click THIS text or scan the QR code for links.](#)



## Term 1

### WEEK 1 15m Natural History: Grade 5 - Lesson 1

#### *My Own Garden*

Materials: Nature All Around: Plants, video links, seeds, pot with soil or place in the ground, trowel, water

PREP: Read Teacher Tip

#### → INTRO

Did you ever think of all the world as a garden full of many different plants? What about plants stands out to you most? Look at the picture on the cover of this book. How many different plants do you see? Do you recognize any of them?

#### → READ, NARRATE, & DISCUSS

p.4-6 "Do you have" - "four basic parts." Take a moment to diagram a plant with these 4 parts.

p.8 "Flowering plants start" - "They shrivel up."

#### → VIEW

Let's watch the beginning of the plant life cycle, and then we'll begin our own gardens, choose one or both based on time:

∞ Video Link: Growing Basil Time Lapse (1:42)

∞ Video Link: Wildflower Garden Time Lapse (3:31)

#### → ACTIVITY

Begin your own small garden in the ground or even in a single pot. Plant seeds according to the seed packet and plan to check on them and water them regularly. The soil should be moist, but not wet. If it seems wet, then don't add more water. If you gently blow across the surface and the soil has a dusty character, then give it some water. As seeds sprout over the next few weeks, make careful observations in your nature journal.

#### → NARRATE & DISCUSS

Use a nature journal to record what you planted.

#### ★ TEACHER TIP

The livingness of this book lies in its artwork, so read the assigned pages, but know that some students may learn more from reading only a part and instead exploring and spending time with the pictures. When students narrate, they are analyzing the "big idea."

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 2 15m Natural History: Grade 5 - Lesson 2

#### *We Depend on Plants*

Materials: Nature All Around: Plants, video links

#### → INTRO

What do you recall from our last reading about plants? Why do you think plants are important to us? Plants make the air we breathe; they shape our communities and traditions; they even make many of our most important medicines. Let's find out how they do the first of these in our book.

#### → READ, NARRATE, & DISCUSS

p.9 "A plant's green leaves" - "even at night."

●Add to last week's diagram or sketch a new one, showing how the plant releases oxygen into the air.

#### → VIEW

∞ Video Link: Ethnobotanist (to 3:03)

∞ Video Link: Life-saving Medicines (to 4:13)

#### → NARRATE & DISCUSS

#### ● OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

# Natural History: Grade 5

[Click THIS text or scan the QR code for links.](#)



## Term 1

### WEEK 3 15m Natural History: Grade 5 - Lesson 3

#### *Getting to Know Flowering Plants*

Materials: Nature All Around: Plants, video link, Shanleya's Quest card game

#### → INTRO

Recall what we learned in the last lesson. What do we call someone who studies plants? What do they actually do?

#### → VIEW

∞ Video Link: What is a Botanist? (6:31)

#### → READ, NARRATE, & DISCUSS

p.10-11 "Take a look" - "from the flowers."

- If you have the Shanleya's Quest card game, then explore the deck today. Notice differences in the 8 plant families presented and how the key characteristics are repeated in different plants in the family. If you do not have the card game, then refer to the link (about halfway down the webpage).

∞ Website Link: Shanleya's Quest Patterns in Plants

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.